

Inside a computer

The earliest computers were simple calculators. At a basic level, computers haven't changed much since then. They take in data (input), perform calculations, and give out answers (output).

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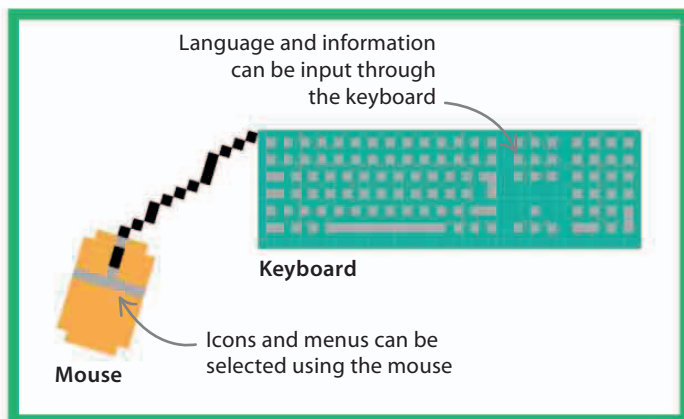
Basic elements

A computer consists of four main parts: input, memory, processor, and output. Input devices gather data, similar to the way your eyes or ears collect information about the world around you. Memory stores the data, while processors examine and alter it, just like a human brain. Output devices show the results of the processor's calculations, like a person speaking or moving after deciding what to do.

▷ Von Neumann architecture

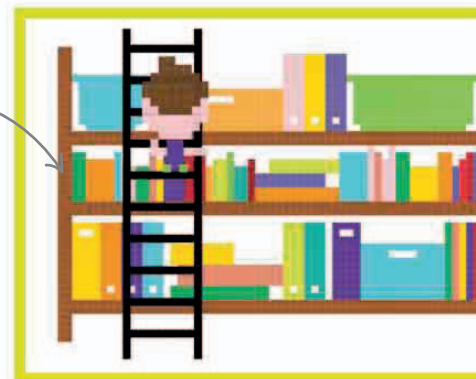
A scientist called John von Neumann first came up with the standard layout for a computer in 1945. His plan is still followed today, with some improvements.

Input



The memory contains information in sections, like books on library shelves. Memory is used to store programs and the data they use

Memory



The control unit loads and carries out instructions from programs

Processor

